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David Suarez
Commissioner
City of Miami Beach
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Dear Commissioner Suarez,

I am writing to you as a resident and citizen of Miami Beach. I submitted a FOIA records request to the City of Miami Beach as well as the Miami-Dade Water and Sewer Department in 2022 for the most recent drinking water testing data for the City of Miami Beach. I was directed to the Annual Water Quality Report.

In no unclear terms, the City of Miami Beach drinking water supply is highly contaminated with PFAS chemicals that are more than 1000x higher than the EPA Health Advisory levels published in 2022.¹ The City of Miami Beach has a duty to provide Miami's residents with safe, clean drinking water. This letter demands action to immediately address the PFAS contamination in our vital drinking water supply by immediately warning the public that they should be using widely available filtration in their homes, schools, etc. (carbon filtration, reverse osmosis, etc.) as well as the City of Miami Beach establishing a plan to rectify the contaminated drinking water supply. Notably, reverse osmosis filtration removes over 99% of PFAS chemicals from the water, which is implemented by the municipality of North Miami Beach for its citizens as well as the Hialeah plant in Miami-Dade.

PFAS Impacts

PFAS chemicals are called "Forever Chemicals" because they do not leave the human body, as they just accumulate forever in the body and they can cause significant health impacts. According to the EPA² these impacts are most dangerous to children and pregnant women, including but not limited to:

¹ <https://www.epa.gov/system/files/documents/2022-06/drinking-water-ha-pfas-factsheet-communities.pdf>

² <https://www.epa.gov/pfas/our-current-understanding-human-health-and-environmental-risks-pfas>

- Reproductive effects such as decreased fertility or increased high blood pressure in pregnant women.
- Developmental effects or delays in children, including low birth weight, accelerated puberty, bone variations, or behavioral changes.
- Increased risk of some cancers, including prostate, kidney, and testicular cancers.
- Reduced ability of the body's immune system to fight infections, including reduced vaccine response.
- Interference with the body's natural hormones.
- Increased cholesterol levels and/or risk of obesity.

Because children are still developing, they may be more sensitive to the harmful effects of chemicals such as PFAS. They can also be exposed more than adults because children drink more water, eat more food, and breathe more air **per pound of body weight than adults**, which increases their exposure to PFAS.

PFAS Exceedances: PFOS and PFOA

The 2023 City of Miami Beach water quality report (published in December 2024) shows that levels of Perfluoro octane sulfonate ("PFOS") and Perfluorooctanoic acid ("PFOA") are substantially higher than the USEPA's established Maximum Contaminant Level (MCL).³

	City of Miami Beach Levels	Miami-Dade MWSD Levels	USEPA Advisory Levels	USEPA Drinking Water Standards (MCL)
PFOS	38 ppt	34 ppt	.02 ppt	4 ppt
PFOA	30 ppt	29 ppt	.004 ppt	4 ppt

In addition, the 2022 City of Miami Beach water quality report shows that levels of Perfluorooctane sulfonate ("PFOS") and Perfluorooctanoic acid ("PFOA") measured at 36 parts per trillion ("ppt") and 12 ppt, respectively. The 2019 - 2021 reports show similar numbers. This is substantially higher than the EPA Health Advisory Level of .02 ppt for PFOS and .004 for PFOA, and the proposed USEPA drinking water standard of 4 ppt.

³ <https://www.epa.gov/system/files/documents/2022-06/drinking-water-ha-pfas-factsheet-communities.pdf>

According to EPA's website, "EPA's lifetime health advisories identify levels to protect all people, including sensitive populations and life stages, from adverse health **effects resulting** from exposure throughout their lives to these PFAS in drinking water."⁴ In other words, detections above the health advisory level pose a health concern.

The City of Miami Beach drinking water exceeds EPA's health advisory by orders of magnitude and is objectively unsafe for human consumption. Yet, the City's website and annual water quality report, which are the only means of communication with their customers, falsely say the water is safe especially for a child to consume. This is alarming.

The City of Miami Beach supplies drinking water to its residents with PFOS and PFOA levels that exceed the EPA advisory levels by 1800x and 3000x, respectively.

I understand that the City of Miami Beach purchases their drinking water from the Miami-Dade Water and Sewer Department ("MWSD") and that MWSD does the testing. MWSD does not perform treatment for PFOS or PFOA, and the City of Miami Beach does not perform treatment for PFOS or PFOA. The City of Miami Beach supplies that water to the residents of Miami beach, including children and pregnant women.

City of Miami Beach Water Quality Reports

The City of Miami Beach, (CMB) Public Works Department publishes information on its water supply and describes the water as follows in its annual report⁵:

- "The Water Quality Report is designed to inform you about the excellent water the City of Miami Beach delivers to you every day."
- The excellent level of our drinking water
- Our number one goal is to provide you and your family a safe and dependable supply of drinking water.
- Our more than 48 employees strive to deliver a quality product and protect the County's precious water resources.
- To ensure the safety of your water, CMB routinely monitors for contaminants in your drinking water according to federal, state, and local laws, rules and regulations.
- CMB delivers drinking water to more than 90 thousand people each day, which, is purchased from Miami Dade County

⁴ <https://www.epa.gov/system/files/documents/2022-06/drinking-water-ha-pfas-factsheet-communities.pdf>

⁵ <https://www.miamibeachfl.gov/city-hall/public-works/operations-division/storm-water-utility/>

- In order to ensure that water is safe to drink, the EPA prescribes regulations, which limit the amount of certain contaminants in water provided by public water systems

Notably, in your annual report you show multiple happy smiling children drinking water likely to convey how safe the water is (below). According to the USEPA, the PFOS and PFOA levels are dangerous and can cause many different problems in children.



For Customers with Special Health Concerns

Some people may be more vulnerable to contaminants in drinking water than the general population. Immuno-compromised persons such as persons with cancer undergoing chemotherapy, persons who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and infants can be particularly at risk from infections. These people should seek advice about drinking water from their health care providers. EPA/CDC guidelines on appropriate means to lessen the risk of infection by Cryptosporidium and other microbiological contaminants are available from the Safe Drinking Water Hotline at 1-800-426-4791.

Para los Consumidores con Preocupaciones Especiales de Salud

Es posible que algunas personas sean más vulnerables a los contaminantes hallados en el agua potable que la población en general. Las personas con sistemas inmunológicos afectados por diversas razones como los pacientes de cáncer que reciben tratamiento de quimioterapia, personas que han recibido un trasplante de órgano, personas con VIH o SIDA u otra enfermedad del sistema inmunológico, algunas personas mayores y bebés, pueden tener mayor riesgo de infección. Estas personas deben consultar el uso de agua potable con su médico. Las pautas de la EPA/CDC sobre las maneras apropiadas de reducir el riesgo de infección por exposición a criptosporidio y otros contaminantes microbiológicos están disponibles a través de Línea Directa del Agua Potable Segura en el 1-800-426-4791.

The contamination is so significant that Miami-Dade County is seeking a \$199 million settlement from 3M, as part of a Multi District Litigation (“MDL”) settlement in which 3M will settle for over \$10 billion for PFOS PFOA contamination to water suppliers nationwide. Miami-Dade County is expected to receive \$199 million over a multi-year period as damages. Miami-Dade formally entered the MDL in 2020 and has not engaged in remediation of PFOA or PFOS, but continues to supply untreated water to their residents. There does not appear to be published plans for treatment in the immediate future by Miami-Dade nor the City of Miami Beach. Treatment appears to be years away.

PFOS and PFOA Health Concern:

PFOS and PFOA are part of a class of industrial chemicals used for various commercial and military operations that have contaminated the City of Miami Beach water supply.

PFOS and PFOA are dangerous chemicals that can cause serious health problems, including cancer, reduced vaccine response, and reduced infectious disease resistance. They are especially dangerous to children and pregnant women due to their ability to bioaccumulate.

PFOS and PFOA are per- and poly-fluoroalkyl substances (PFAS) which are a large and diverse group of chemicals used in many commercial applications due to their unique properties, such as resistance to high and low temperatures, resistance to degradation, and nonstick characteristics. Although PFAS have been manufactured and used broadly in commerce since the 1940s, concern over potential adverse effects on human health grew in the early 2000s with the detection of PFOA and PFOS in human blood. Since that time, hundreds of different PFAS have been found in water, soil, and air. Many PFAS are made up of long chains of carbon-fluorine bonds, such as PFOA and PFOS. They are environmentally persistent, bioaccumulative, and remain in human bodies for a long time. The more PFOA and PFOS a person is exposed to, the more it accumulates in the body.

According to the EPA, human studies have show PFOA and/or PFOS exposure may result in several types of negative health effects including:

- Cancer
- Immune (suppression of vaccine response)
- Cardiovascular
- Liver
- Human development, such as decreased birthweight

Furthermore, a sensitive non-cancer effect is suppression of vaccine response (decreased serum antibody concentrations) and reduced infectious disease resistance. This is especially dangerous in children and pregnant women and their susceptibility to infectious disease, including reduced vaccine immunity.

This letter serves to describe the impacts on humans but it should be noted the environmental impact is quite significant. Not only is the contaminated water exposing people to PFOA/S via ingestion, but hundreds of millions, if not billions of gallons of contaminated water has gone down drains further contaminating the environment due to the City's failure to obtain an alternative water source or place treatment in their distribution system.

Proven Solutions to Treat Drinking Water for PFOS PFOA

Notably, there is proven technology to treat the water at both a city wide levels as well as the household level. This includes carbon treatment as well as reverse osmosis

(plus remineralization). As previously stated, reverse osmosis is implemented by local miami drinking water municipalities.

Carbon filtration is a method commonly used to treat PFAS contaminated water for entire homes as well as municipalities. In this process, water passes through activated carbon, which has a high surface area and can adsorb PFAS molecules. As the water flows through the carbon filter, the PFAS contaminants adhere to the surface of the carbon, effectively removing them from the water. The result is cleaner, safer water with reduced levels of PFAS. Carbon water filtration is readily available in the market and is most efficient for individual homes entire home water system and municipal water systems.

Reverse osmosis is a water purification process that can effectively remove PFAS (per- and polyfluoroalkyl substances) from contaminated water. In this process, contaminated water is forced through a semipermeable membrane, which blocks the PFAS molecules while allowing clean water to pass through. As a result, the PFAS contaminants are separated from the water, producing purified water that meets safety standards. Reverse Osmosis water filtration is readily available in the market and is most efficient for individual homes drinking water.

What will the City of Miami Beach do about this public health threat?

The EPA advisory levels and proposed drinking water standards are to protect ALL PEOPLE, including sensitive populations and life stages. Notably, the PFOA and PFAS levels detected in the City of Miami Beach drinking water that is supplied to its residents is significantly more than 1000x the EPA advisory levels. It is of the utmost importance to protect the sensitive populations including children and pregnant women, amongst others.

The City of Miami Beach must warn the public immediately, remove children happily drinking water from its water quality report and promote water filtration in addition to planning treatment, which will require funding. The City of Miami Beach should be proceeding with its own plan for treatment, because while the City of Miami Beach may be getting water from the Miami-Dade, the City of Miami Beach is ultimately responsible for the water it supplies to its residents.

The public should be notified immediately about the dangers of PFAS and PFOA as well as the individual, single home treatment options that are readily available to provide clean water (carbon, reverse osmosis). Plans should be commenced to provide clean water to the residents of Miami Beach that is free from PFAS chemicals.

Sincerely,

Gregory J. Rigano